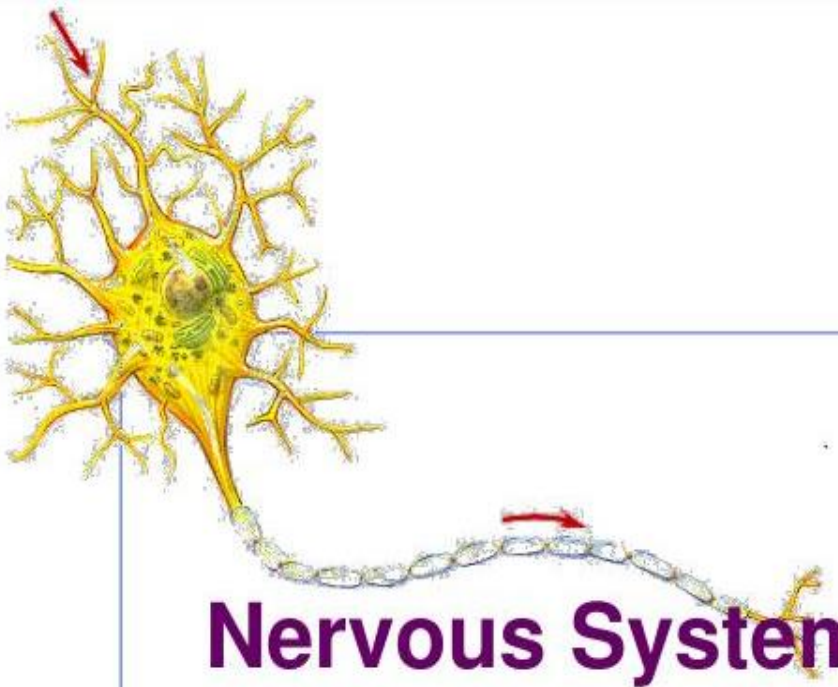
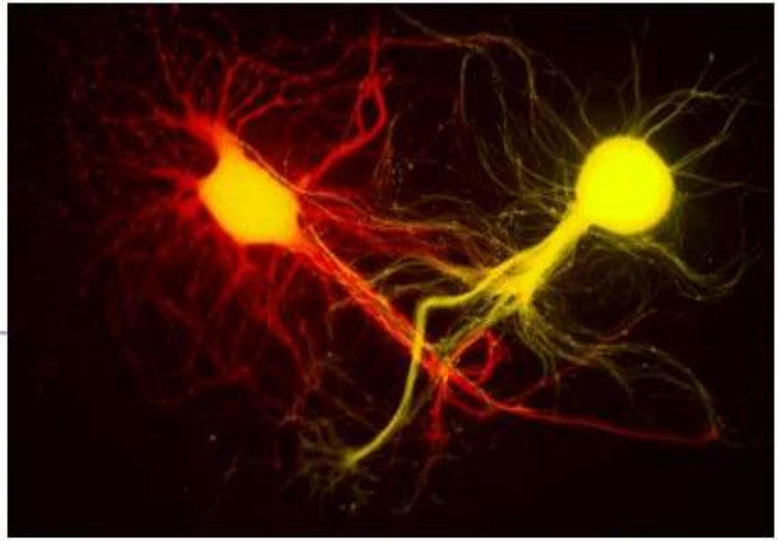
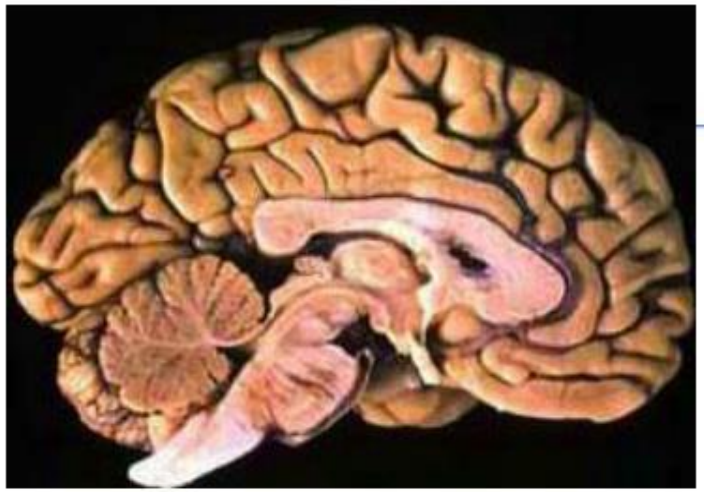
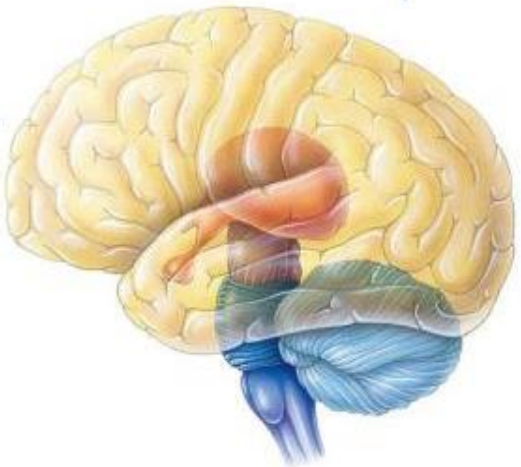




Nervous System: The Neuron and the Transmission of a Nerve Impulse

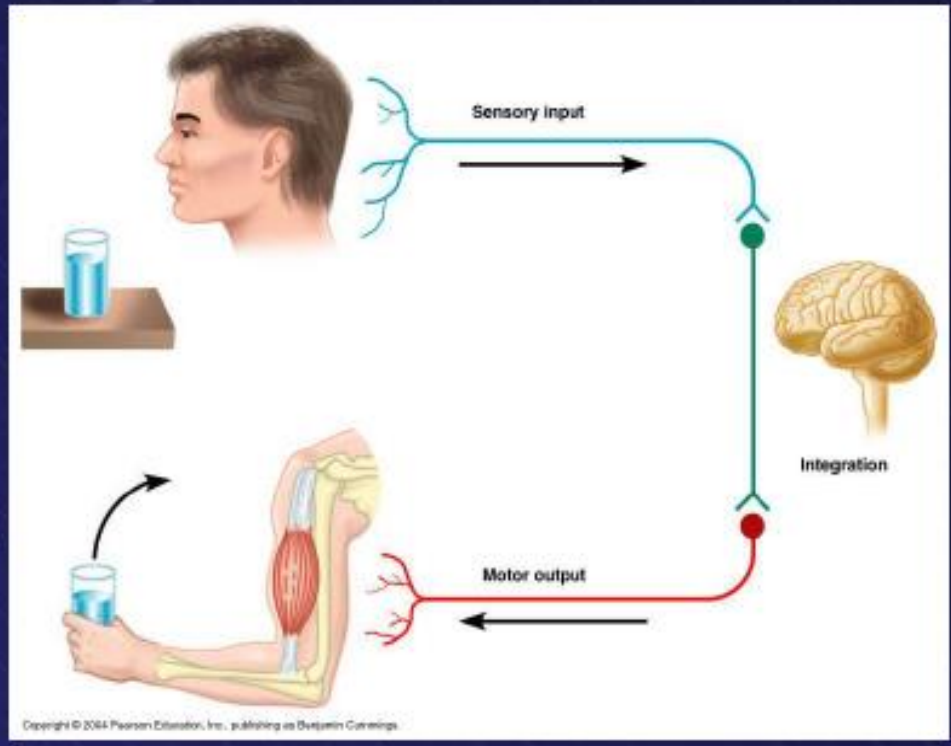






NERVOUS FUNCTIONS

- Body's master controlling and communicating system
- Three functions
  - Sensory input
    - Gathers information from sensory receptors
  - Integration
    - Processes and interprets sensory input
  - Motor output
    - Activates effector organs to cause a response



Functions of the Nervous System

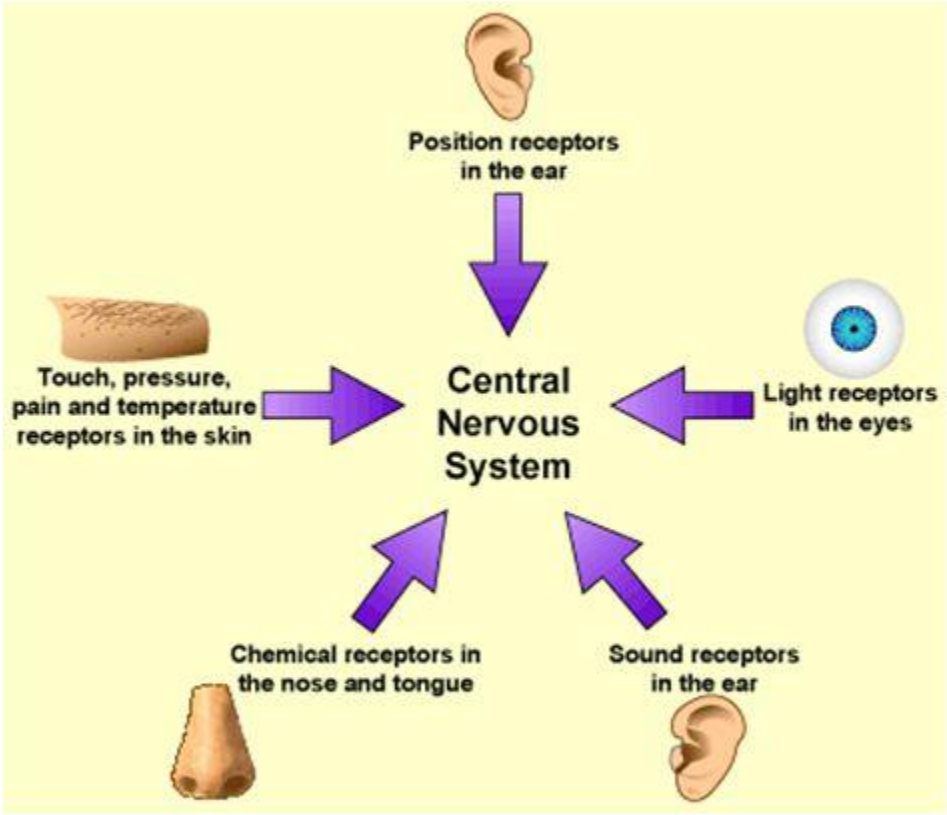
1. Sensory input – gathering information
   - To monitor changes occurring inside and outside the body (changes = stimuli)
2. Integration –
   - to process and interpret sensory input and decide if action is needed.
3. Motor output
   - A response to integrated stimuli
   - The response activates muscles or glands

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Sensory Receptors

Types of sensory **receptors**

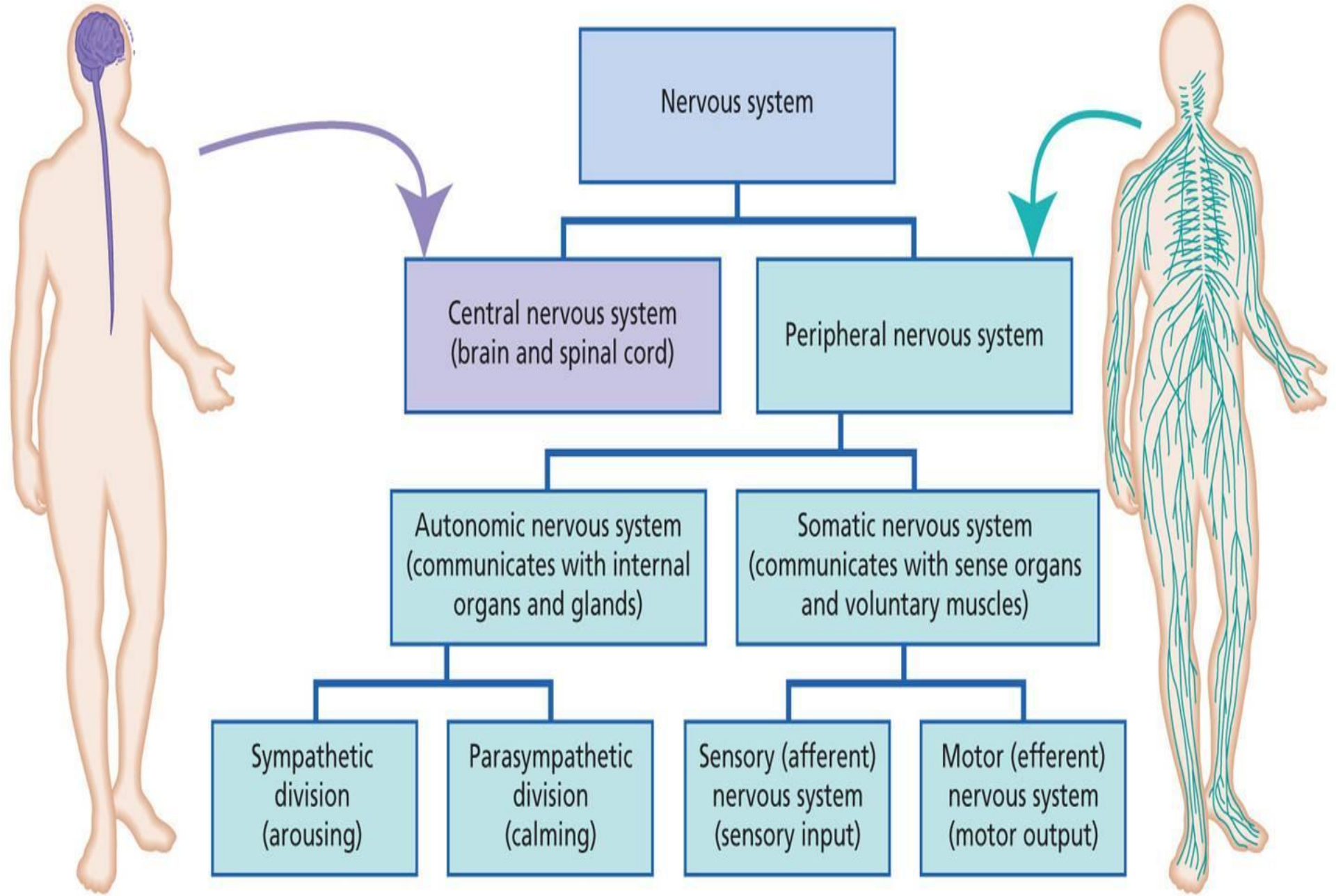
- Touch
- Pain
- Heat
- Cold
- Pressure
- Smell
- Taste



Types of sensory **organs**

- Eyes
- Ears
- Nose
- Skin
- Tongue

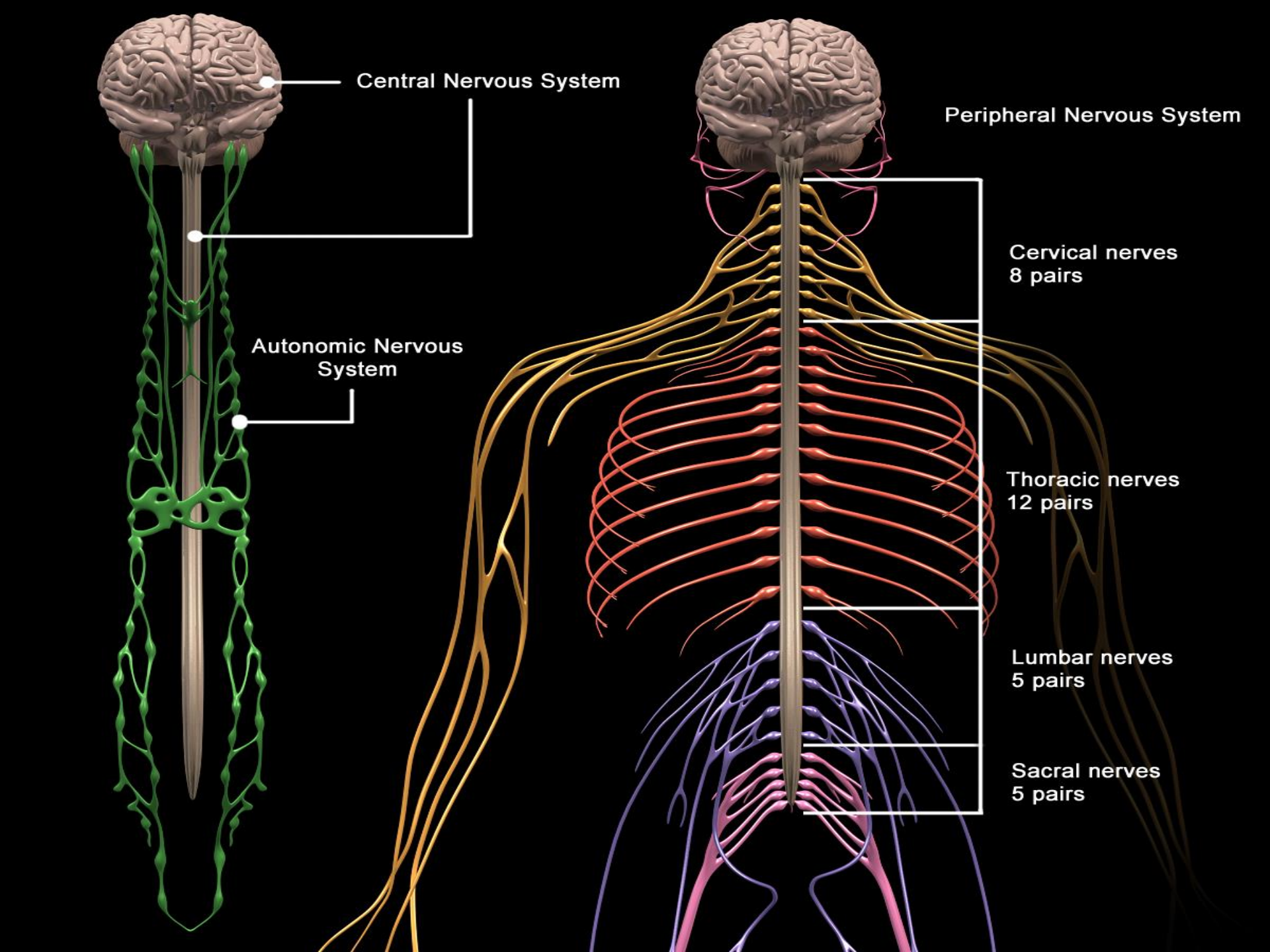
Organization of Nervous system



Structural Classification of the Nervous System

- Central nervous system (CNS)
  - Brain
  - Spinal cord
- Peripheral nervous system (PNS)
  - Nerve outside the brain and spinal cord

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Central Nervous System
Autonomic Nervous System
Peripheral Nervous System
Cervical nerves
8 pairs
Thoracic nerves
12 pairs
Lumbar nerves
5 pairs
Sacral nerves
5 pairs

Central Nervous System: Processes different kinds of incoming sensory information.

Peripheral Nervous System: Includes all nervous tissue outside the CNS. Components of the PNS include nerves, ganglia, and sensory receptors.

Somatic Nervous System:

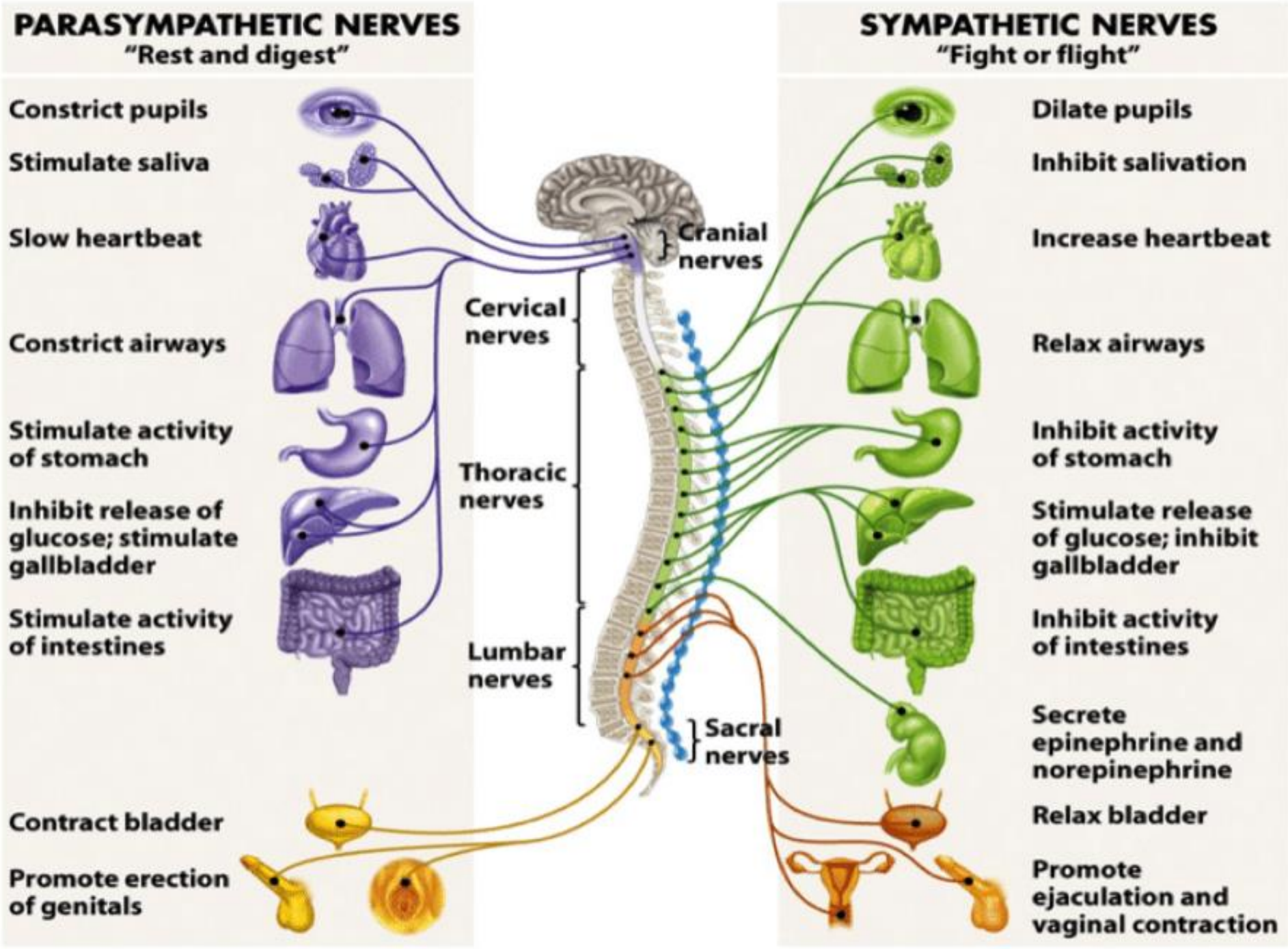
a) **Sensory neurons** that convey information to the Central nervous system from somatic receptors in the head, body wall, and limbs and from receptors for the special senses of vision, hearing, taste, and smell.
b) **Motor neurons** that conduct nerve impulses from the CNS to skeletal muscles only. Because these motor responses can be consciously controlled, the action of this part of the PNS is voluntary.

Autonomic Nervous System:

a) **Sensory neurons** that convey information to the CNS from autonomic sensory receptors, located primarily in visceral organs such as the stomach and lungs.

b) Motor neurons that conduct nerve impulses from the CNS to smooth muscle, cardiac muscle, and glands. Because its motor responses are not normally under conscious control, the action of the ANS is involuntary.

The motor part of ANS consists of two branches, the sympathetic division and parasympathetic division. With few exceptions, effectors receive nerves from both divisions, and usually the two divisions have opposing actions.


PARASYMPATHETIC NERVES
"Rest and digest"
Constrict pupils
Stimulate saliva
Slow heartbeat
Constrict airways
Stimulate activity of stomach
Inhibit release of glucose; stimulate gallbladder
Stimulate activity of intestines
Contract bladder
Promote erection of genitals
SYMPATHETIC NERVES
"Fight or flight"
Dilate pupils
Inhibit salivation
Increase heartbeat
Relax airways
Inhibit activity of stomach
Stimulate release of glucose; inhibit gallbladder
Inhibit activity of intestines
Secrete epinephrine and norepinephrine
Relax bladder
Promote ejaculation and vaginal contraction
Cranial nerves
Cervical nerves
Thoracic nerves
Lumbar nerves
Sacral nerves

You come across a bear on your walk to school…what happens?

- Sympathetic nervous system does what?
- Increases heart rate
- Increases breathing rate
- Dilates bronchioles
- Dilates pupils
- Inhibits digestion



Look at the chart and say the COLOUR not the word.



Left – Right Conflict

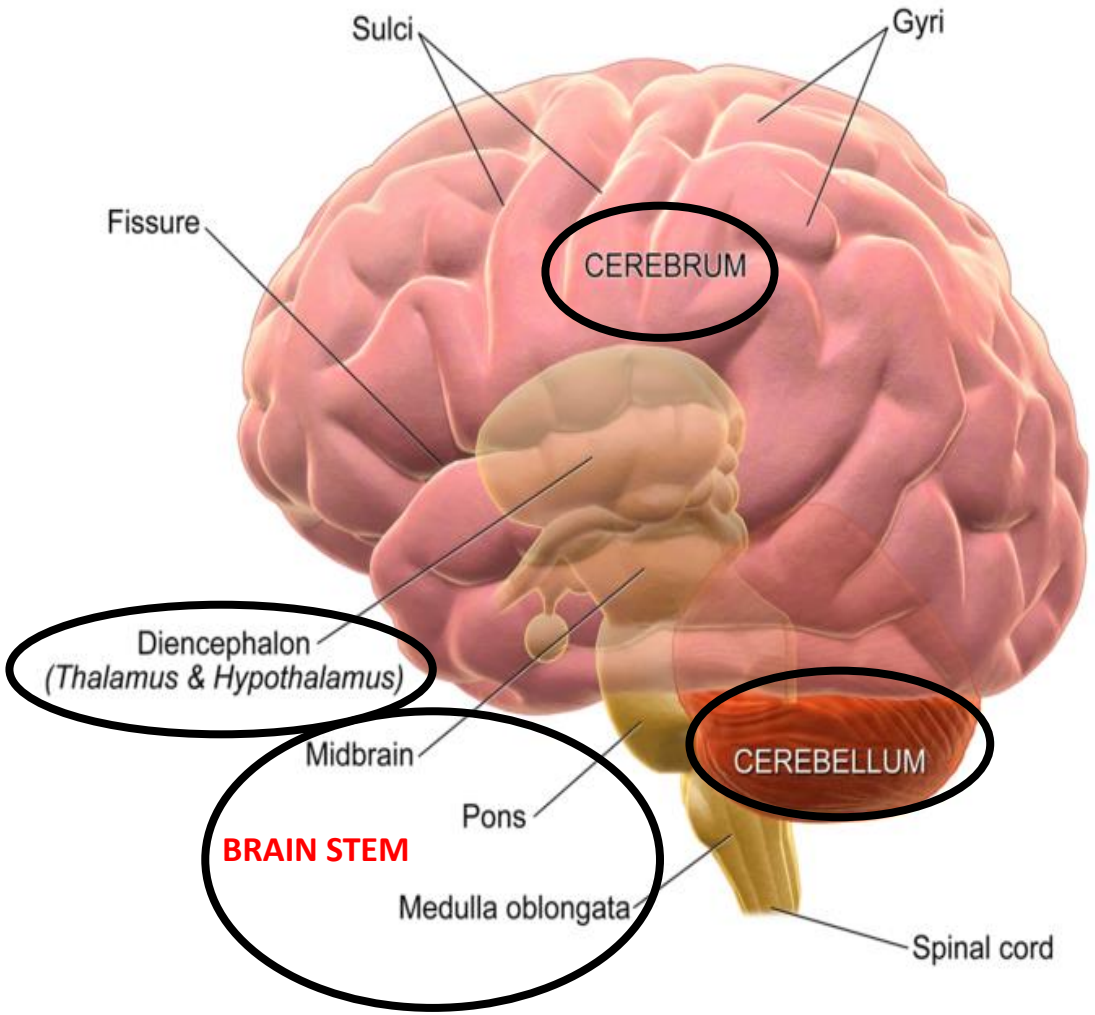
Your right brain tries to say the coulour but your left brain insists on reading the word.

Major subdivisions of human brain

- The brain is one of the largest organs of the body, consisting of about 100 billion neurons and 10-15 trillion neuroglia with a mass of about 1300 g

The four major parts of human brain:

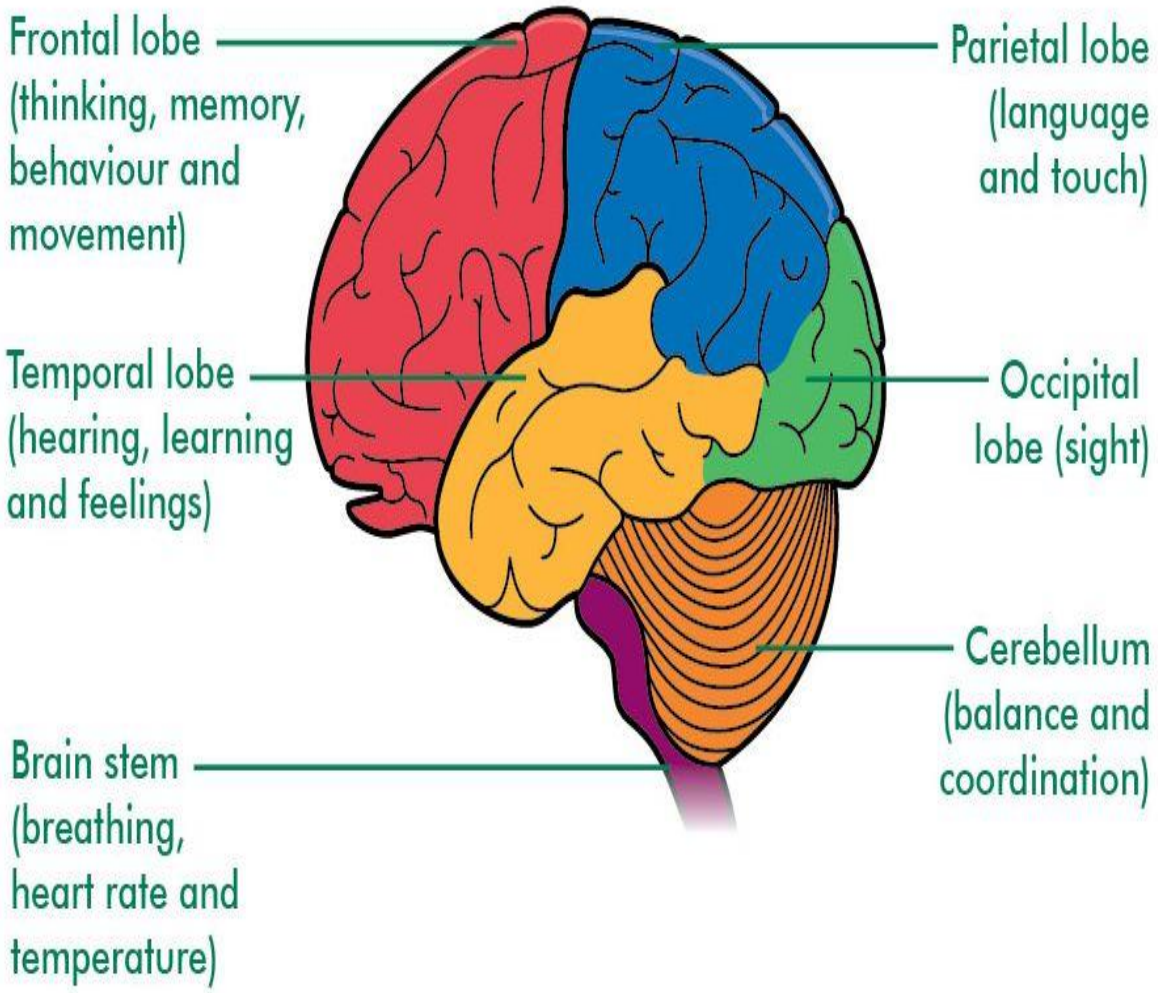
1. **Cerebrum/cerebral hemisphere**
2. **Diencephalon: Thalamus, hypothalamus and epithalamus (pineal gland)**
3. **Cerebellum**
4. **Brain stem: Midbrain, Pons and Medulla oblongata**



CEREBRUM is the largest part of the human brain. It is divided into four lobes: Occipital Lobe, Parietal Lobe, Temporal Lobe and Frontal Lobe

Cerebrum

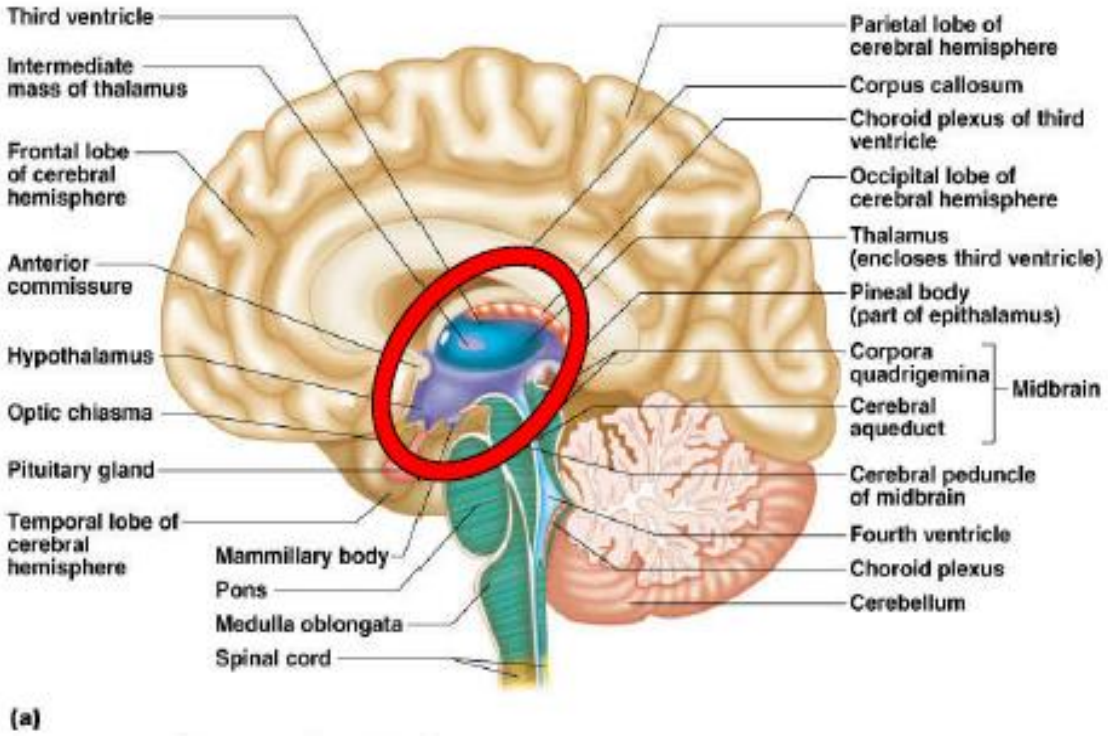
- Frontal Lobe
- Parietal Lobe
- Occipital Lobe
- Temporal Lobe



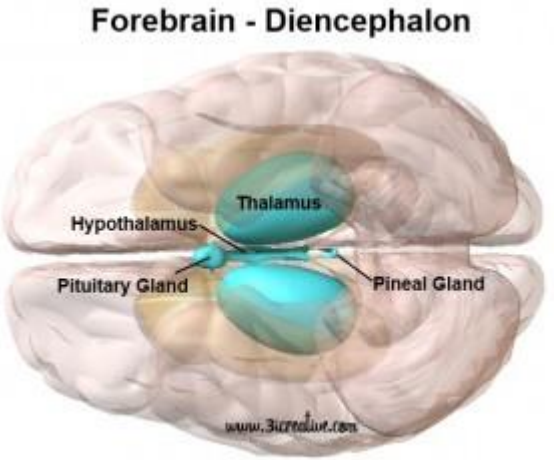
- The surface is made of ridges **(gyri)** and grooves **(sulci)**

Diencephalon

- Sits on top of the brain stem
- Enclosed by the cerebral hemispheres
- Three parts:
  - **Thalamus**
  - **Hypothalamus**
  - **Epithalamus**

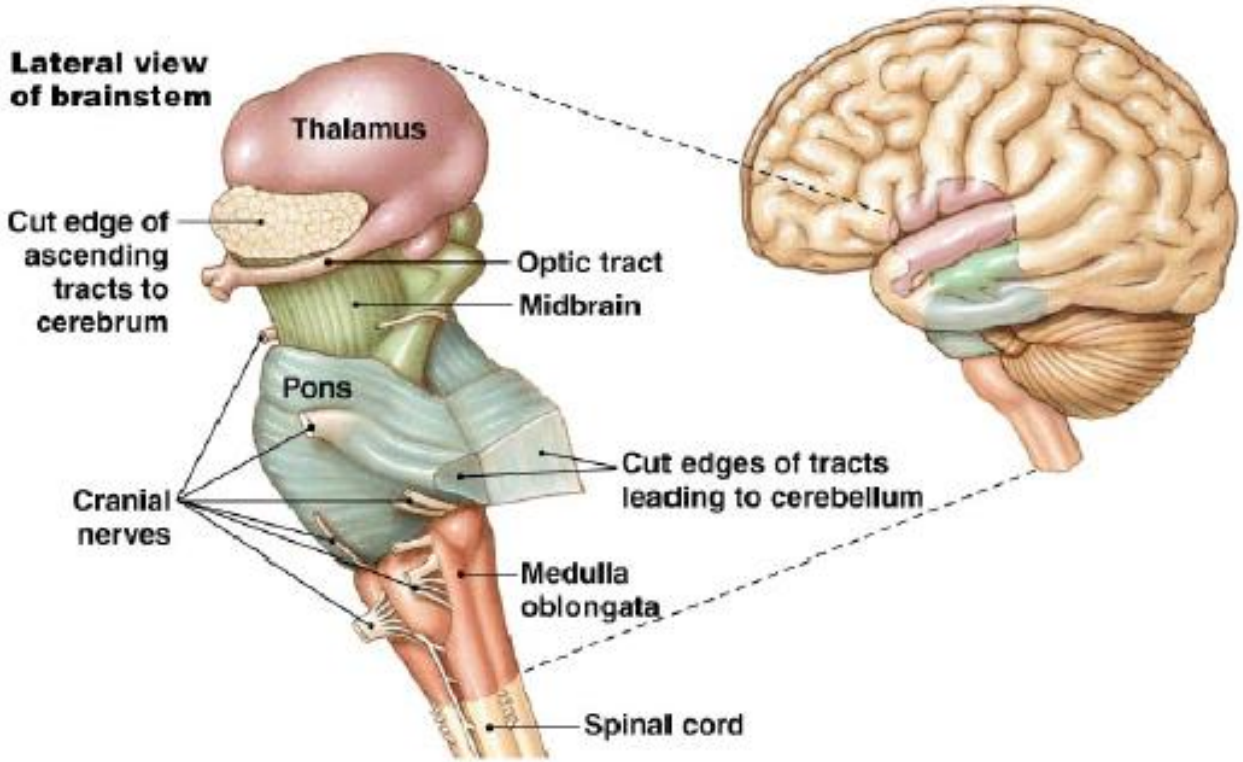


- **Thalamus** is olive shaped structure which performs sensory functions. It directs sensory input from the eyes, ears, tongue and from other sense organs.
- **Hypothalamus** is located below the thalamus as the name suggests. It is important in monitoring hormone concentrations, water concentrations, temperature of the body etc.
- **Epithalamus:** is in the interior of the brain. It is a small area that is primarily covered by the **pineal gland**.
- **Pineal gland** is about a size of small pea. It secretes the hormone melatonin, a very powerful antioxidant that promotes sleepiness, and contributes to the setting of the body's biological clock



Brain Stem

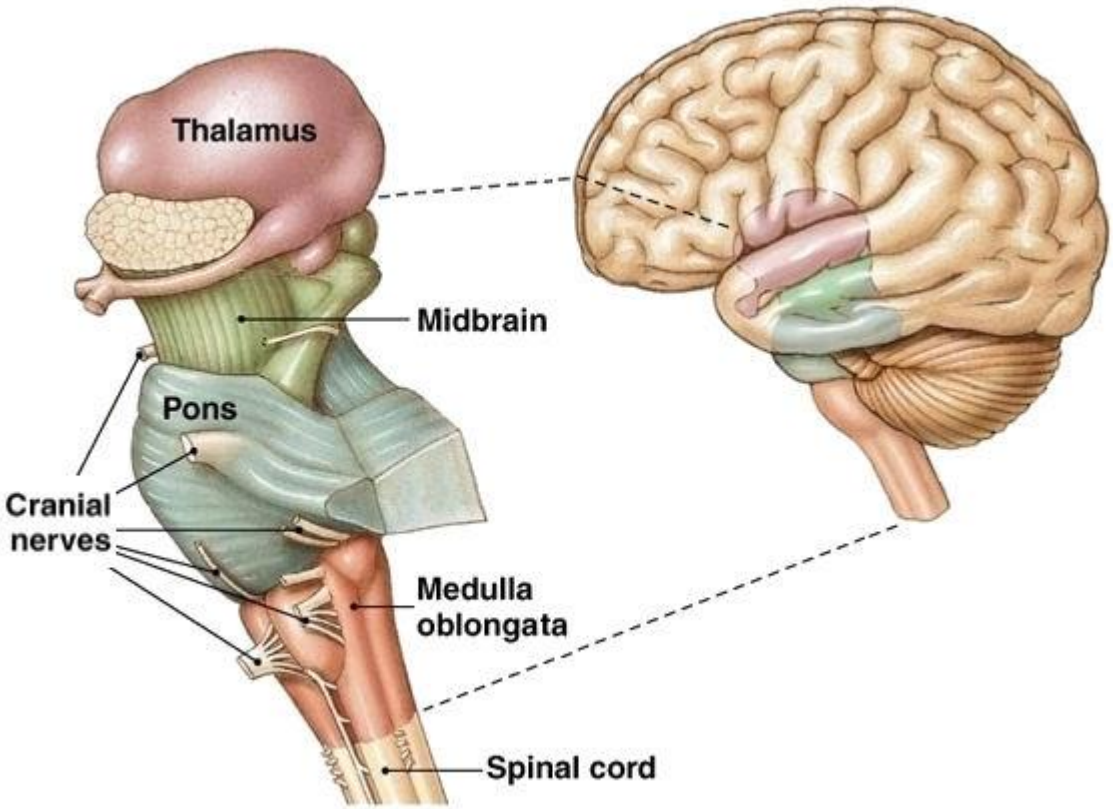
- Attaches to the spinal cord
- Parts of the brain stem:
  - **Midbrain**
  - **Pons**
  - **Medulla oblongata**



Midbrain: It is a small part of the brain. It is associated with Vison, Hearing, Sleep, Arousal, Temperature etc.

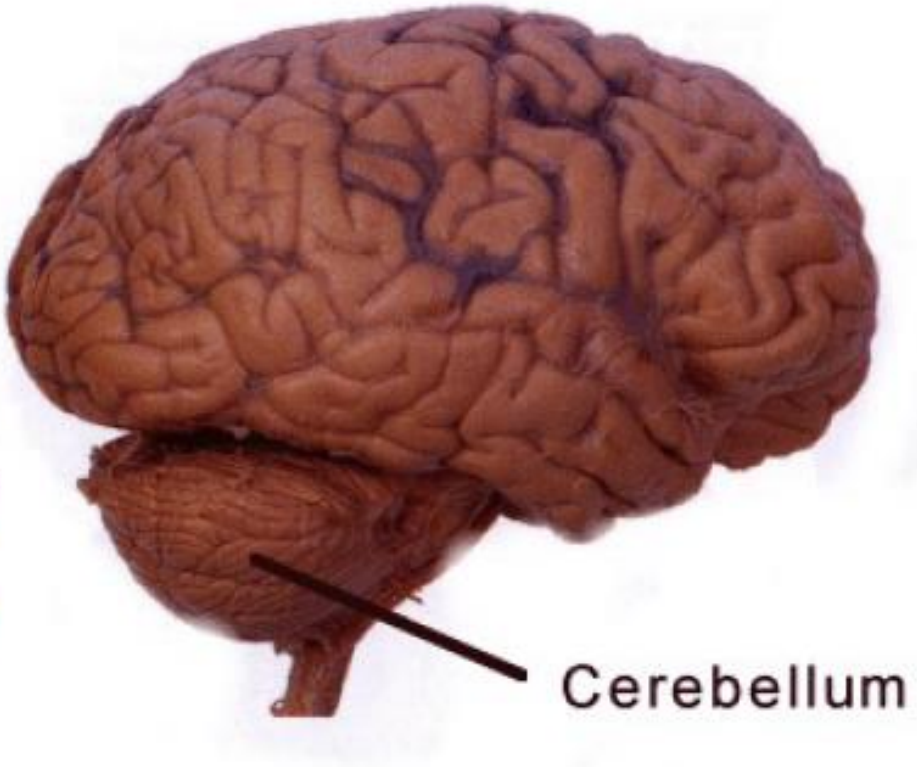
Pons controls digestive and respiratory movements.

Medulla Oblongata connects skull with the spinal cord. It helps in controlling Blood Pressure, Heart Rate, Respiratory Movement, Vomiting etc.



Cerebellum

- Two hemispheres with convoluted surfaces
- Provides involuntary coordination of body movements
- "Arbor vitae" design of white & grey matter



❖ **Cerebellum is involved in controlling attention, language, equilibrium, posture, etc.**

Protection of the Central Nervous System

- Cerebrospinal fluid
- Blood brain barrier

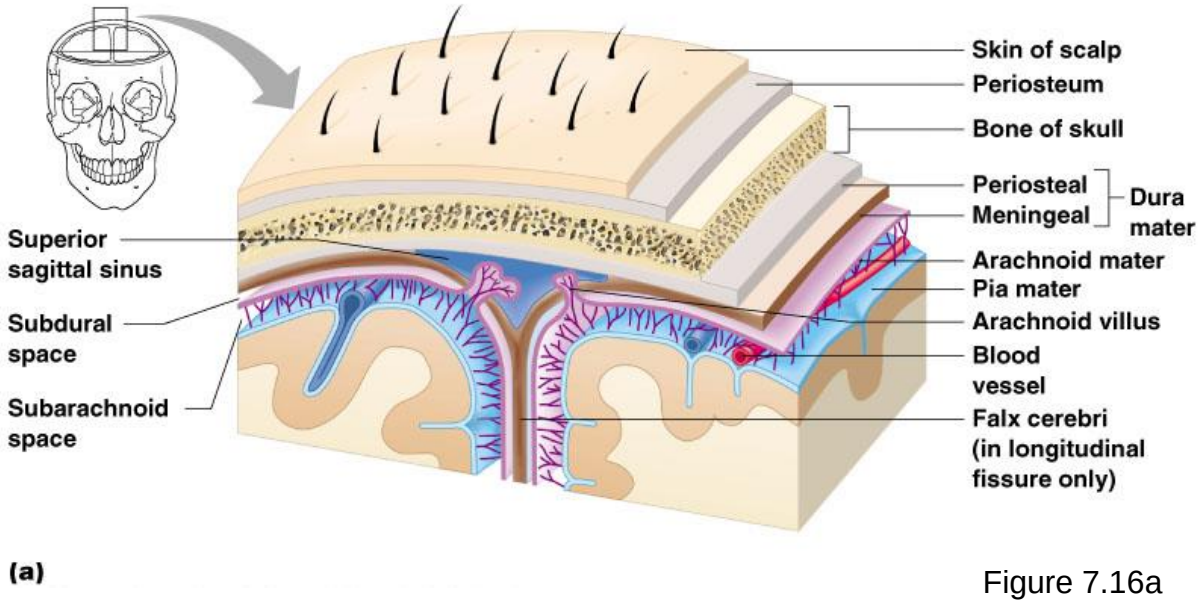
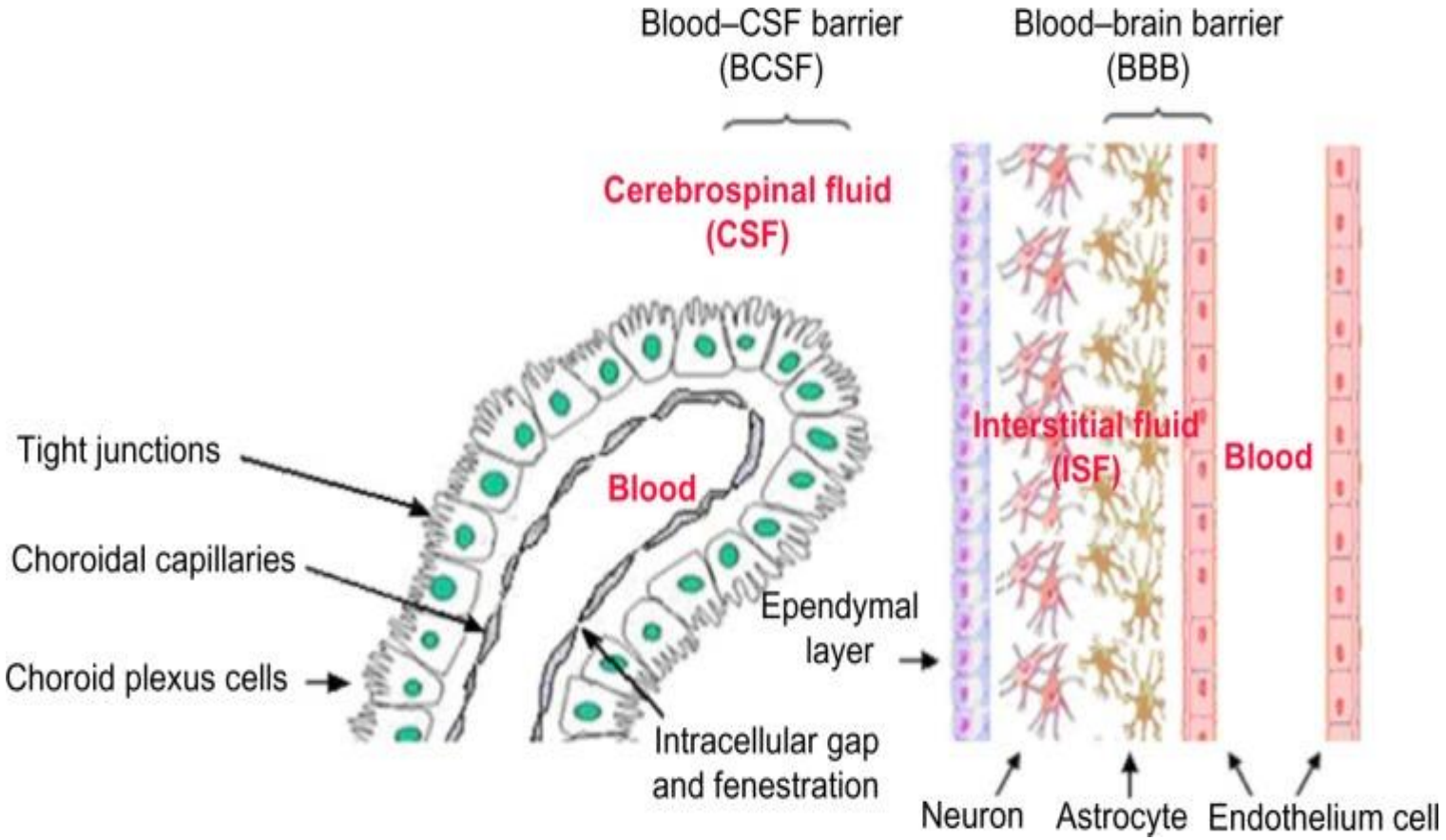


Figure 7.16a

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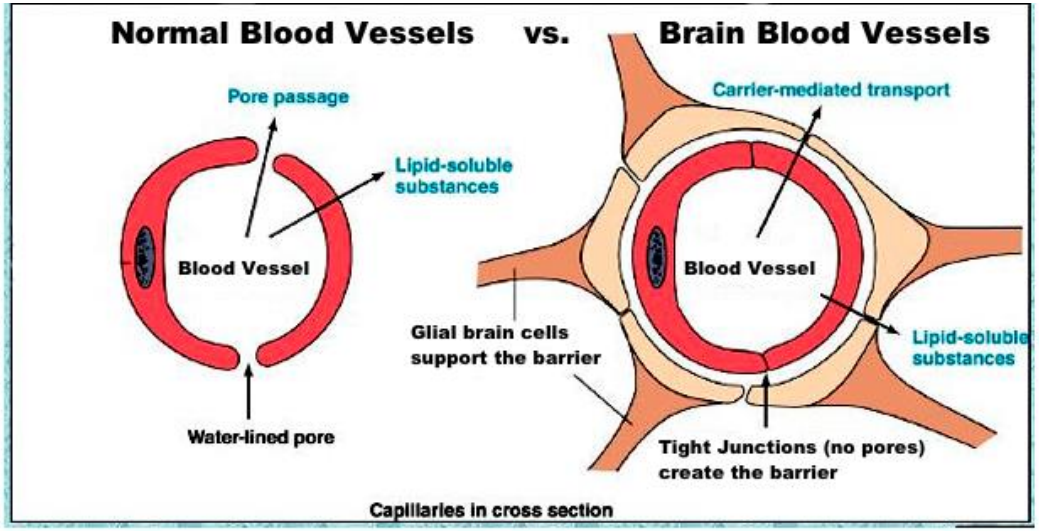
Cerebrospinal fluid (CSF)

CSF is clear, colorless liquid that carries oxygen, glucose, and other needed chemicals from the blood to neurons and neuroglia and removes wastes and toxic substances produced by brain and spinal cord.



The Blood Brain Barrier

- This barrier consists basically of very tightly sealed blood capillaries in the brain assisted by astrocytes.
- The existence of blood-brain barrier protects brain cells from harmful substances and pathogens by preventing passage of many substances from blood into brain tissue.
- Lipid-soluble substances such as oxygen, carbon dioxide, alcohol, and most anesthetic agents easily cross the blood –brain barrier. Trauma, certain toxins and inflammation can cause the breakdown of blood-brain barrier.
- Glucose is the main source of energy for brain cells. Because virtually no glucose is stored in the brain, the supply of glucose must be continuous. Low level of glucose in the blood supply to brain may cause mental confusion, dizziness, convulsions, and loss of consciousness.



WHAT ARE NERVE CELLS?

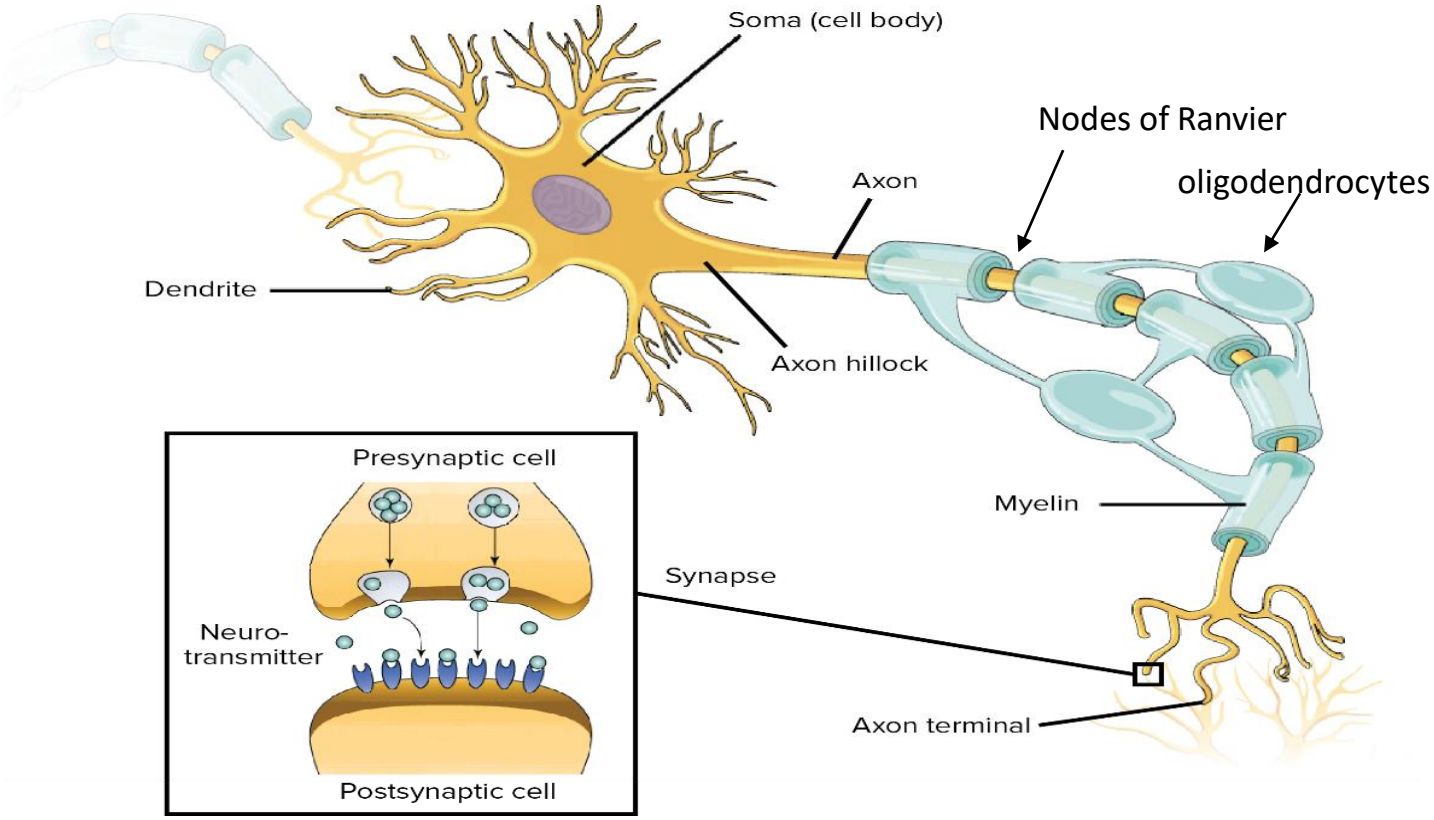
- Neurons are similar to other cells in the body
  - Surrounded by cell membrane
  - Have a nucleus that contains genes
  - Contain cytoplasm, mitochondria and other organelles
  - Carry out basic cellular processes such as protein synthesis and ATP production
- Neurons are different by
  - Specialized extensions called dendrites and axons
  - Communicate with each other by electrochemical process
  - Contain some specialized structures (synapses) and chemicals (neurotransmitters)

Neurons: functional unit of nervous system

The basic components of neuron are similar wherever they occur in the nervous system.

It includes:

- Cell body
- Dendrites
- Axon hillock
- Axon
- Myelin Sheath
- Oligodendrocyte
- Nodes of Ranvier
- Axon terminal
- Synaptic knob

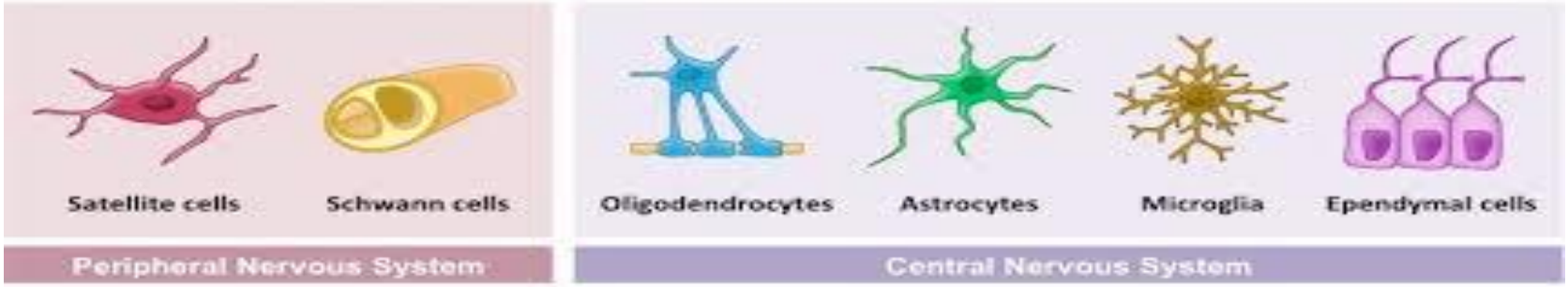


Description of a typical neuron parts

Cell body /Soma	Contains Cytoplasm with typical cell organelles and certain granules. Cell bodies form the **gray matter** of the nervous system. Groups of cell bodies are called **nuclei** in CNS and **ganglia** in PNS. The cell body also contains the chemical machinery to produce the neurotransmitters that the neuron uses to communicate with each other
Dendrites	Cell body plasma membrane extension into short branched fibers. Receive signals from other neurons and sensory receptors.
Axon hillock	Region of the neuron where the cell body narrows to form the axon. Nerve signals are generated here and passes down the axon.
Axon	The neuron's largest and thinnest projection. The nerve signal travels from cell body along the axon to the synapse.
Axon terminal	End of axon which may be single or branched.
Synaptic knob	A bulb-like structure present at the end of axon terminal branches. Synaptic knob possesses synaptic vesicles containing chemicals called neurotransmitters.
Myelin sheath	Wrap-around or covering that insulates the axon and speeds signal conduction. **Made by oligodendrocytes in CNS and Schwann cells in PNS.** Myelinated nerve fibers are found in spinal and cranial nerves. Unmyelinated nerve fibers are found in autonomous and somatic neural systems
Nodes of Ranvier	Gaps between two adjacent myelin sheath

Neuroglia- cells of nervous tissues that hold them together and support neurons

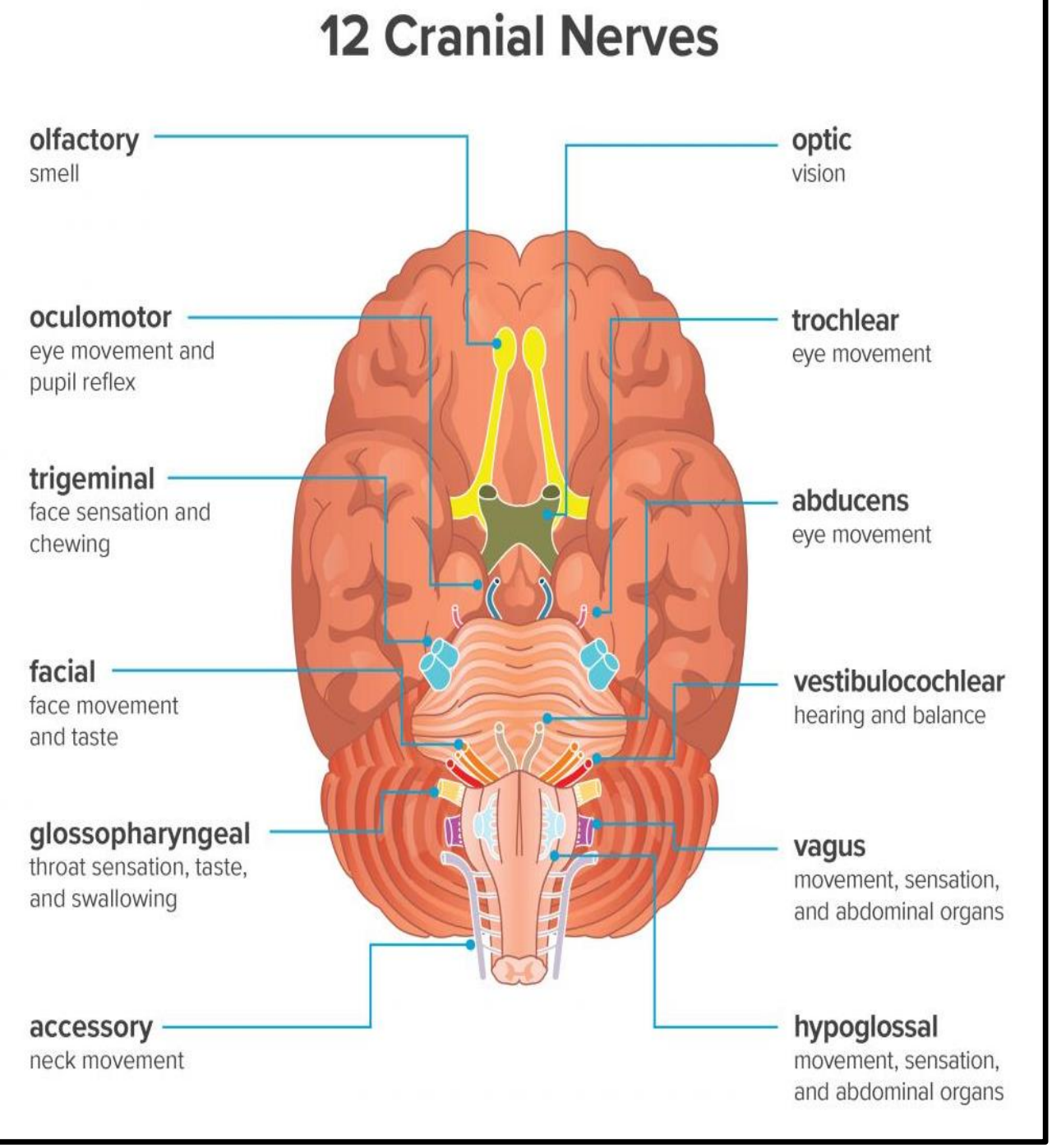
Oligodendrocytes	Produce and maintain myelin sheath around several adjacent axons of CNS.
Astrocytes	Provides neuron with physical support and nourishment in CNS, help maintain the chemical environment for generation of nerve impulses, assist with growth and migration of neurons during brain development.
Microglia	Protects CNS from disease by engulfing invading microbes; migrate to areas of injured nerve tissue where they clear debris of dead cells
Ependymal cells	Form cerebrospinal fluid and assist in its circulation. Line ventricles of brain and central canal of the spinal cord
Schwann cells	Produce and maintain myelin sheath around a single axon of a PNS neuron, participate in regeneration of PNS axons
Satellite cells	Support neurons in PNS ganglia and regulate exchange of materials between neurons and interstitial fluid.



Types of Neurons

- **Sensory neurons**
  - Carry nerve impulses from a receptor to the CNS
  - Have long dendrites and short axons
- **Motor neurons**
  - Carry nerve impulses from the CNS to an effector (ex. muscle or gland)
  - Have short dendrites and long axons
- **Interneurons (Mixed)**
  - Found completely within the CNS
  - Provide a link within the CNS between sensory neurons and motor neurons
  - Have short dendrites and long or short axons

12 Cranial Nerves



The cranial nerves are 12 pairs of nerves that can be seen on the ventral (bottom) surface of the brain. Some of these nerves bring information from the sense organs to the brain; other cranial nerves control muscles; other cranial nerves are connected to glands or internal organs such as the heart and lungs.

Cranial Nerve Name

- I - Olfactory
- II - Optic
- III - Oculomotor
- IV - Trochlear
- V - Trigeminal
- VI - Abducens
- VII - Facial
- VIII- Vestibulocochlear
- IX - Glossopharyngeal
- X - Vagus
- XI - Spinal Accessory
- XII - Hypoglossal

Cranial nerves

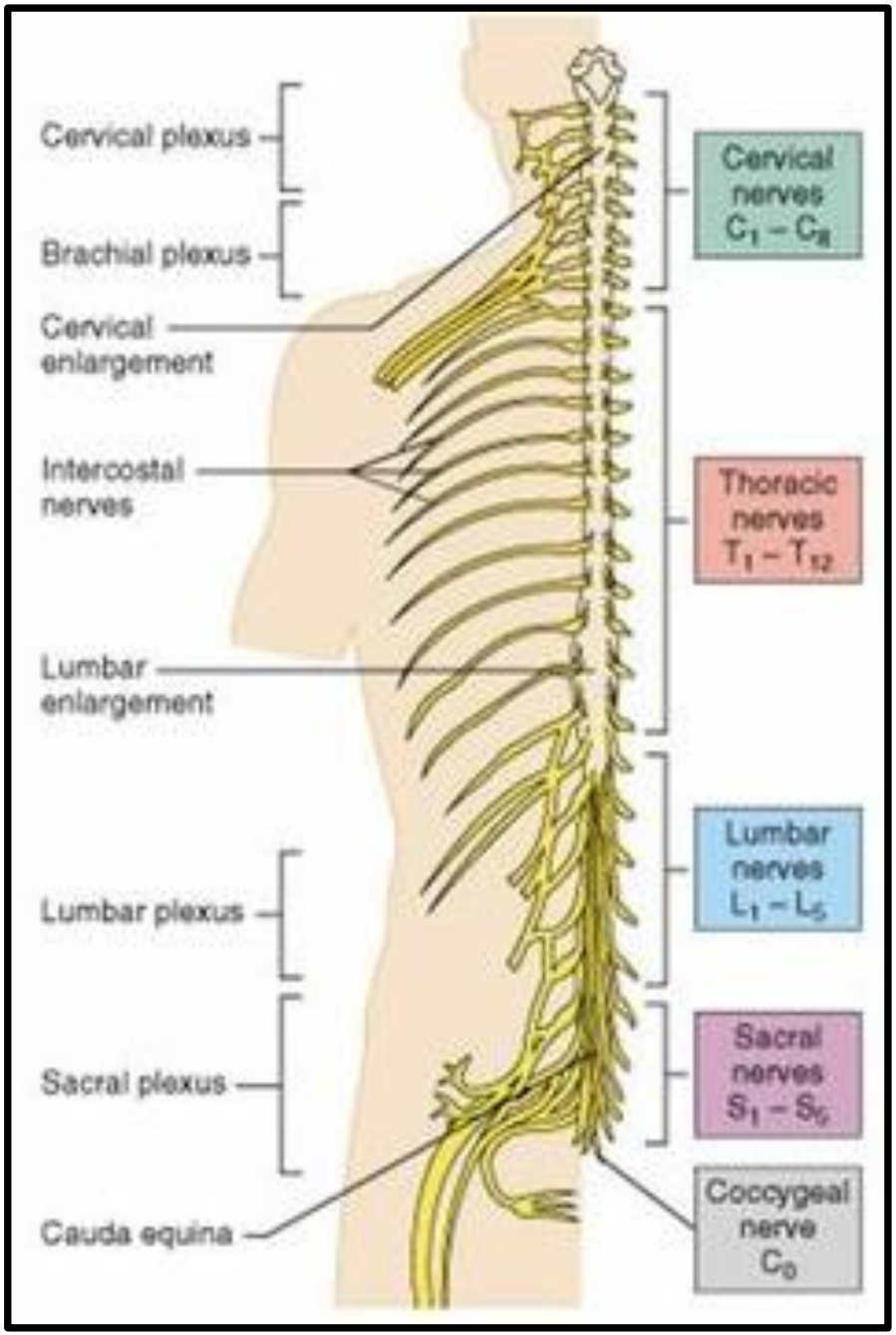
Cranial nerves		What type of nerve
I	Olfactory	Sensory nerve (information conveyed to CNS)
II	Optic nerve	Sensory
III	Oculomotor	Motor nerve (information from CNS to effector cells)
IV	Trochlear	Motor
V	Trigeminal	**Mixed nerve (sensory and motor part)**
VI	Abducens	Motor
VII	Facial	**Mixed**
VIII	Vestibulocochlear	Sensory
IX	Glossopharyngeal	**Mixed**
X	Vagus nerve	**Mixed**
XI	Accessory	Motor
XII	Hypoglossal	Motor

Spinal nerves

31 pairs of spinal nerves forms the path of communication between spinal cord and specific body regions

Spinal nerves:

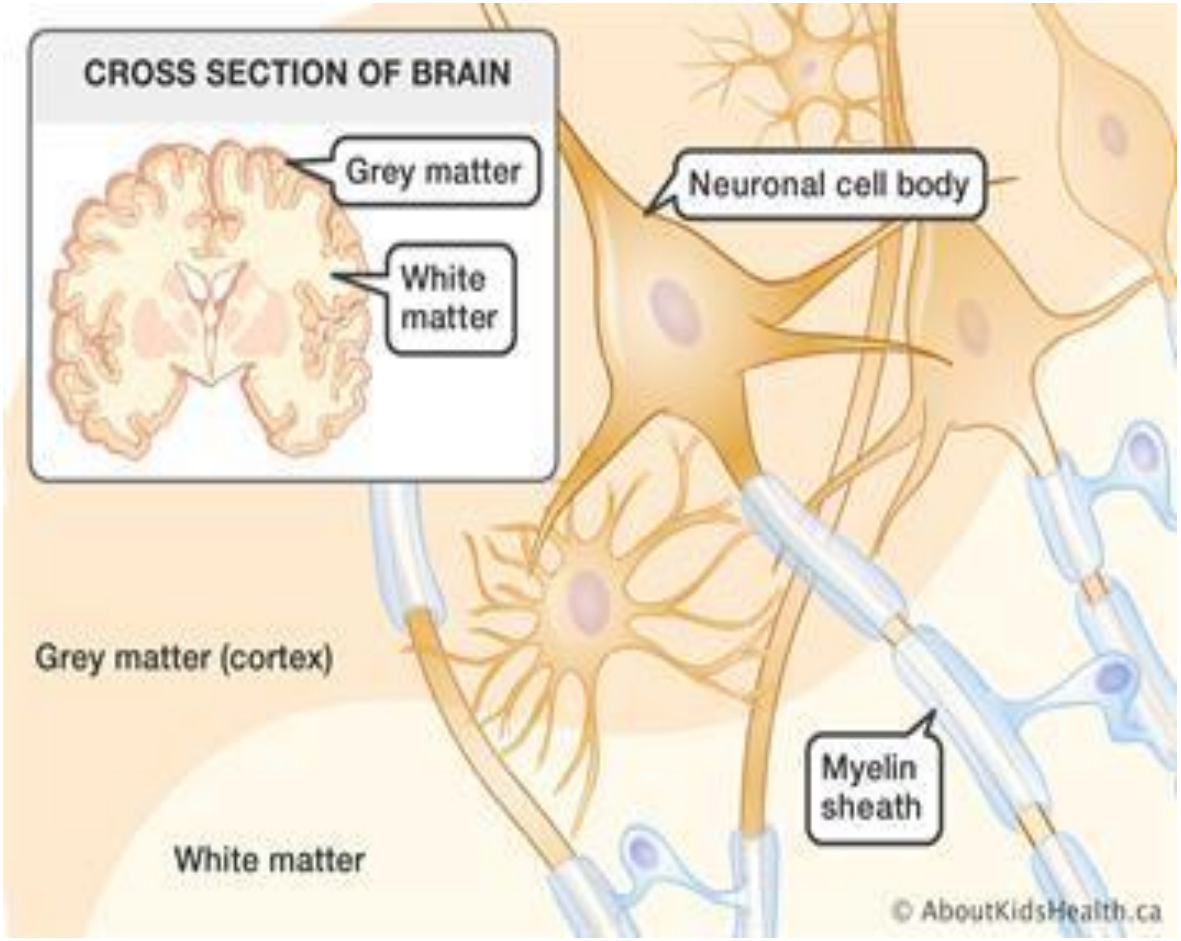
1. **8 pairs of cervical spinal nerves**
2. **12 pairs of thoracic spinal nerves**
3. **5 pairs of lumbar spinal nerves.**
4. **5 pairs of sacral spinal nerves**
5. **1 pairs of coccyx spinal nerves.**



What is gray and white matter of nervous system

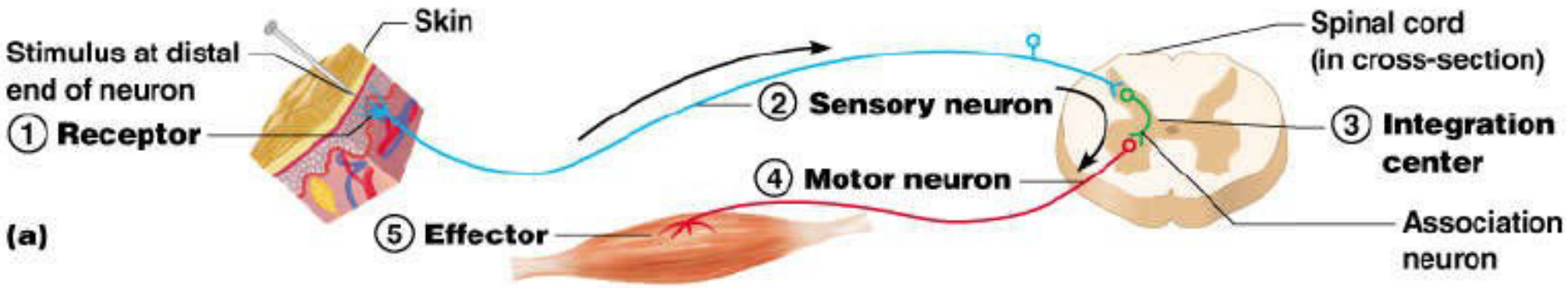
- **Gray matter** of the nervous system contains neuronal cell bodies, dendrites, unmyelinated axons, axon terminals and neuroglia. There is little or no myelin in these areas.
- **White matter** is composed primarily of myelinated axons. In spinal cord dissected section, the outer white matter surrounds the inner core of gray matter

- **Gray matter contains most of the brain's neuronal cell bodies**
- **Makes up about 10%-20% of brain content**
- **Gray matter: speech, hearing, vision, memory, feeling, intelligence**
- **White matter: heart rate, breathing, temperature. etc.**

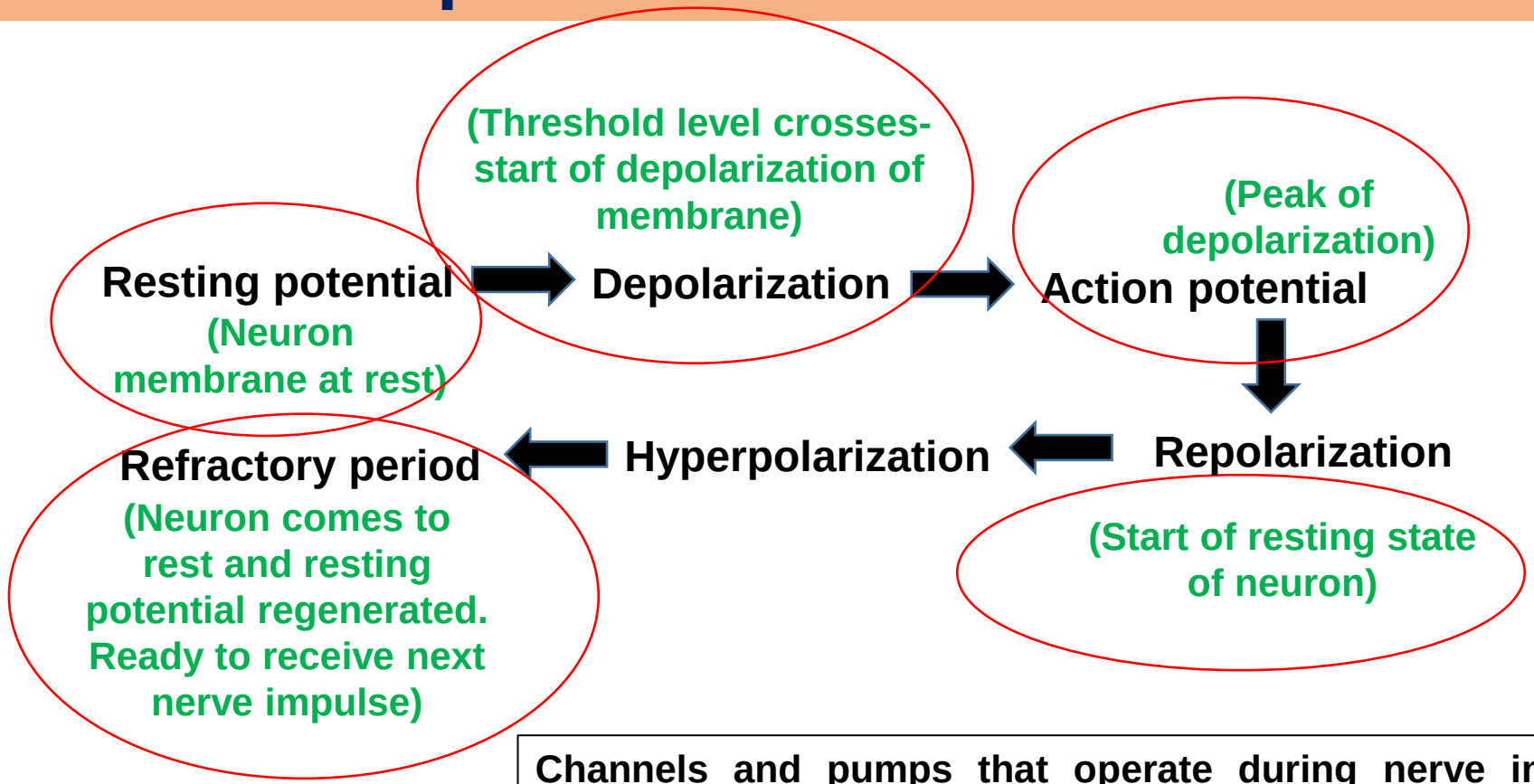


The Reflex Arc

- **Reflex** – rapid, predictable, and involuntary responses to stimuli
- **Reflex arc** – direct route from a sensory neuron, to an interneuron, to an effector



Nerve impulse transmission in neuron



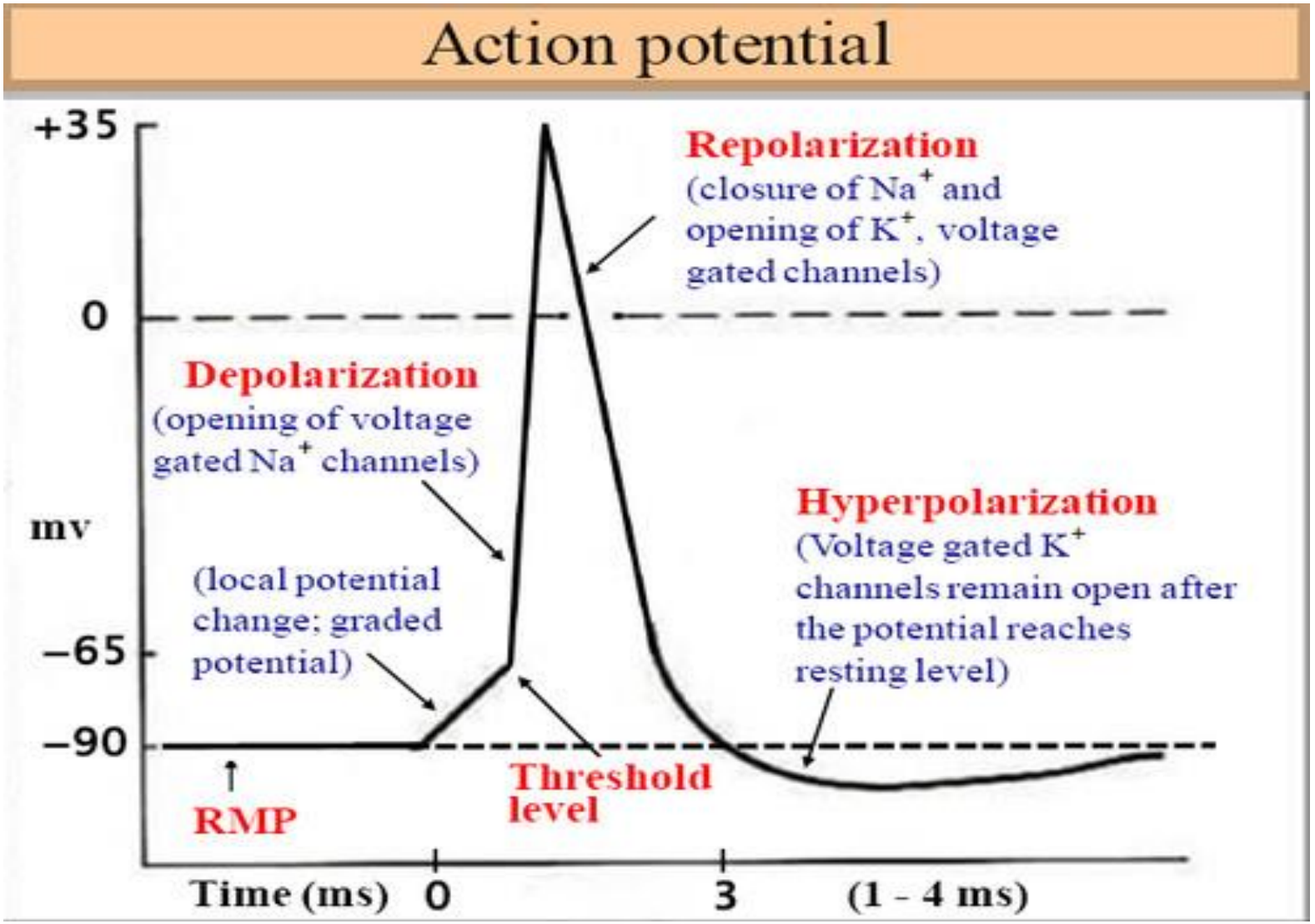
Channels and pumps that operate during nerve impulse transmission:

1) **Voltage gated –potassium channels:** voltage sensor, allow passive flow of K^+ ions across the axon membrane.

2) **Voltage gated –Sodium channels:** voltage sensor, allow passive flow of Na^+ ions across the axon membrane.

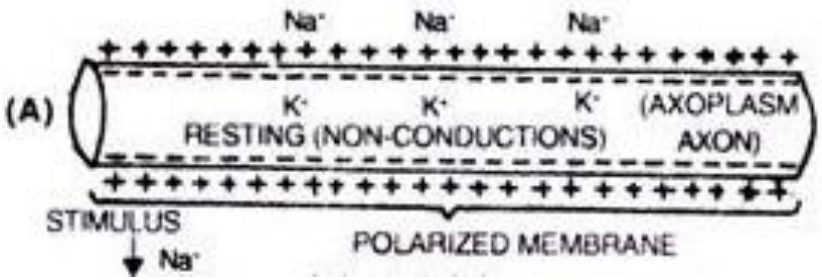
3) **Sodium-potassium pumps:** active transporter, uses ATP to transport $3Na^+$ towards the outside of axon membrane and $2K^+$ ions are released inside the axon membrane.

Graphical representation of nerve impulse transmission

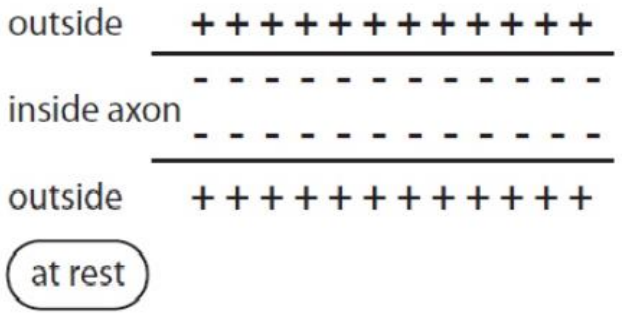


Description of the events

1. **Polarization and resting potential-** membrane is polarized due to the differential distribution of charges across the membrane. **The outer side of the axonal membrane is positive** and **inner side of axonal membrane is negative**. This is due to more sodium ions with respect to less potassium ions inside which is maintained by the sodium-potassium active pumps. This pump throws out $3Na^+$ from inside (axoplasm) to outside (extracellular fluid ECF) in exchange of $2K^+$ ions from ECF to the axoplasm. The **resting membrane potential is recorded -70 to -90 millivolts** (minus 70mV to 90mV). **Sodium channel is closed, potassium channel is closed, and sodium-potassium pump is active**



- When the neuron is not sending an impulse, the charge difference is maintained inside and outside of the axon cell. This is called the resting potential where it is polarized.

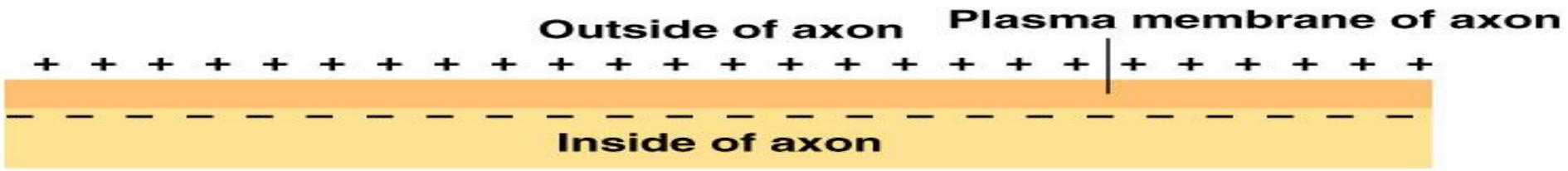


2. Depolarization and action potential- After getting stimulus, membrane is depolarized. Outside of the membrane becomes negative and inside of the axon membrane becomes positive because of entry of sodium ions through sodium ion channel from ECF towards axoplasm. **Potassium ion channel is closed, and sodium-potassium pump is inactive.**

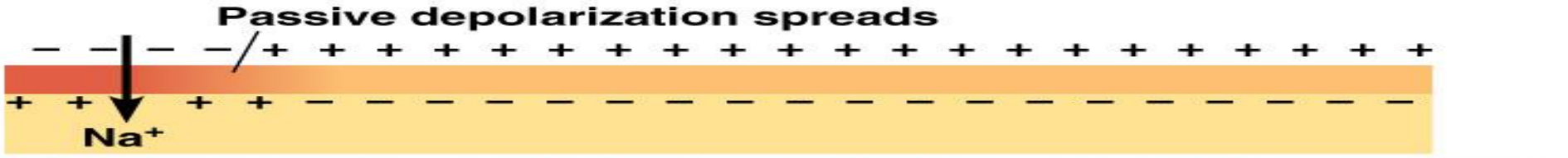
The action potential is the state when all the sodium channels are open and axonal membrane is completely at depolarized state. In other words, it is the peak of complete depolarization of the membrane that is inside positive and outside negative. Or moving towards depolarization is called action potential. **The action potential of the membrane is recorded as +30mV.**

3. Repolarization.

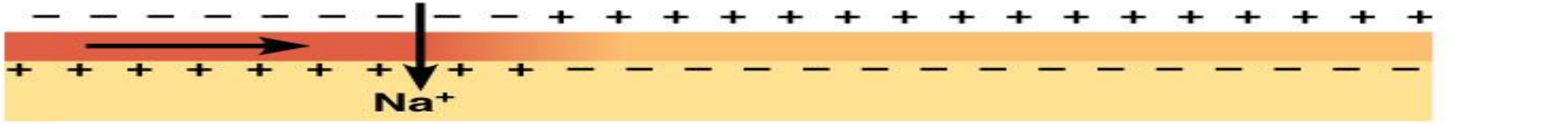
Following depolarization, the sodium ion channels that had been opened while the cell was undergoing depolarization, closes again. The increased positive charge within the cell now causes the potassium ion channels to open. Potassium ions (K^+) begin to move down the electrochemical gradient. As potassium ion moves out of the cell the potential within the cell plummets and approaches its resting potential once more.



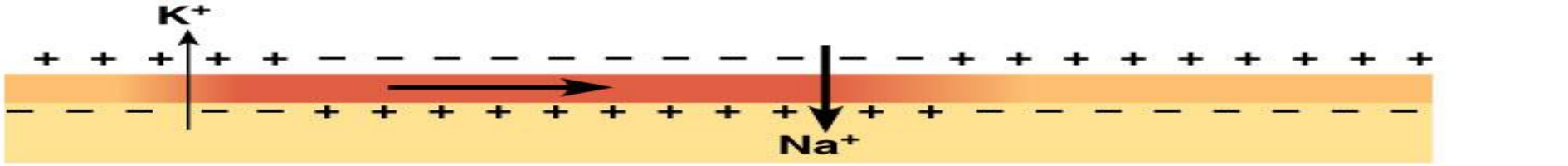
1 At the start, the membrane is completely polarized.



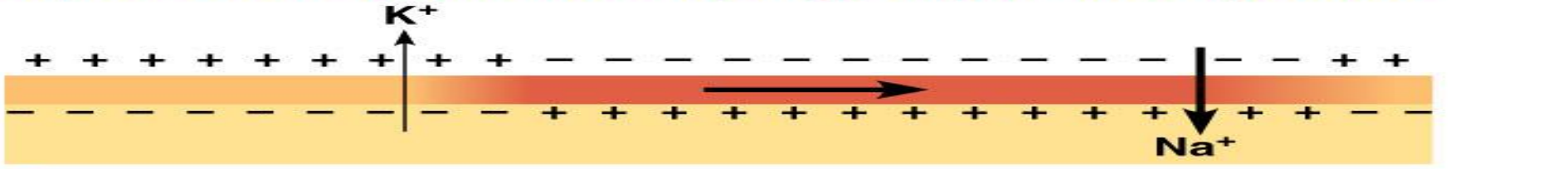
2 When an action potential is initiated, a region of the membrane depolarizes. As a result, the adjacent regions become depolarized.



3 When the adjacent region is depolarized to its threshold, an action potential starts there.



4 Repolarization occurs due to the outward flow of K^+ ions. The depolarization spreads forward, triggering an action potential.



5 Depolarization spreads forward, repeating the process.

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Nerve impulse conduction in myelinated and non-myelinated nerve fibers

Impulse Conduction

Myelinated axons conduct impulses differently than unmyelinated axons.

Unmeylinated Axon generates a series of action potentials along the entire axon

The impulse is slow (travels at 1 mile/hour)

Myelinated Axon

Myelin is an electrical insulator and prevents action potentials along myelinated portions of the axon.

Action potentials are generated only at the Nodes of Ranvier

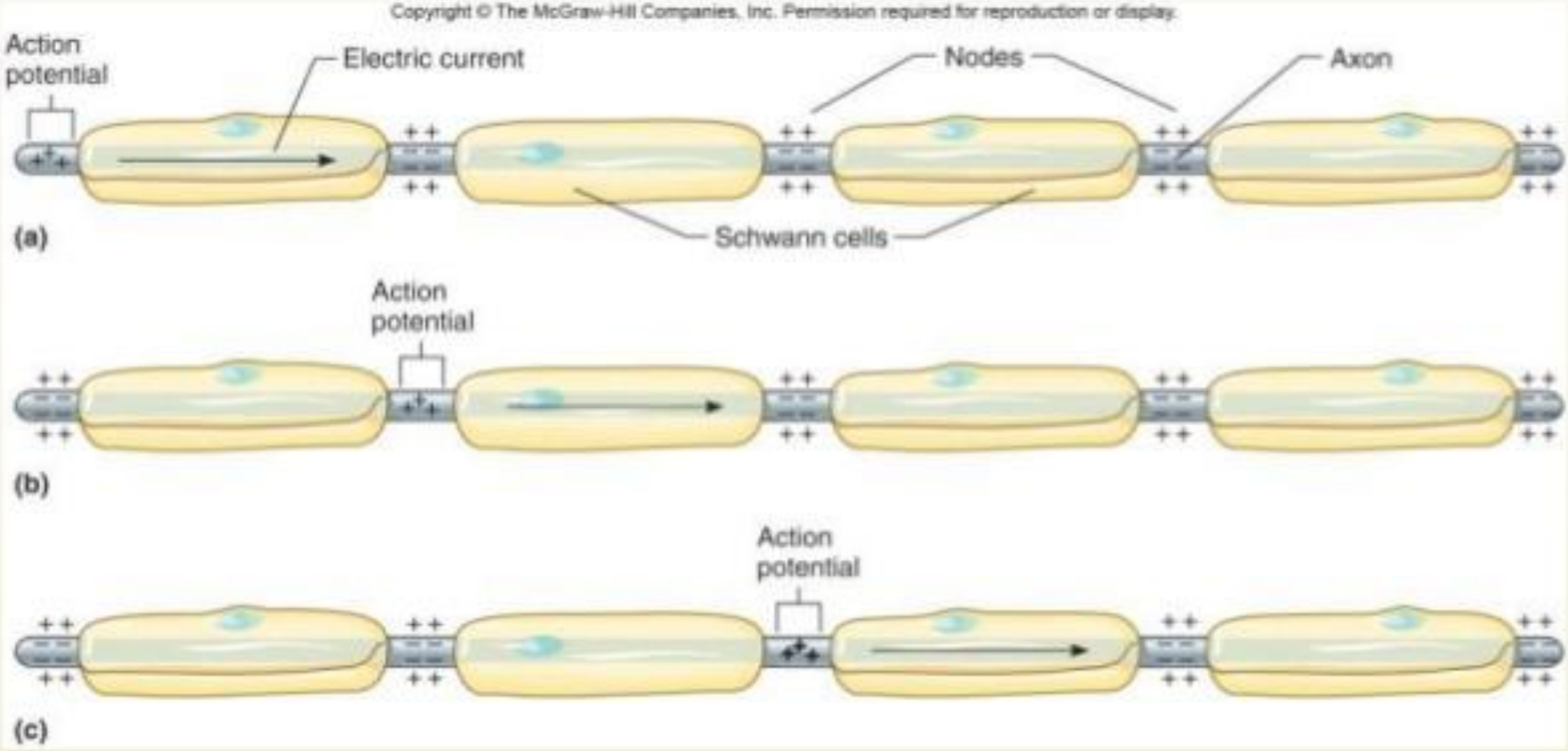
The impulse travels through the myelinated portions by **electrical conduction.**

The impulse is fast (travels at 285 miles/hour!)

Nerve impulse conduction in myelinated nerve fibers: saltatory conduction

Myelinated fibers: Cranial (brain) and spinal (spinal cord) nerves

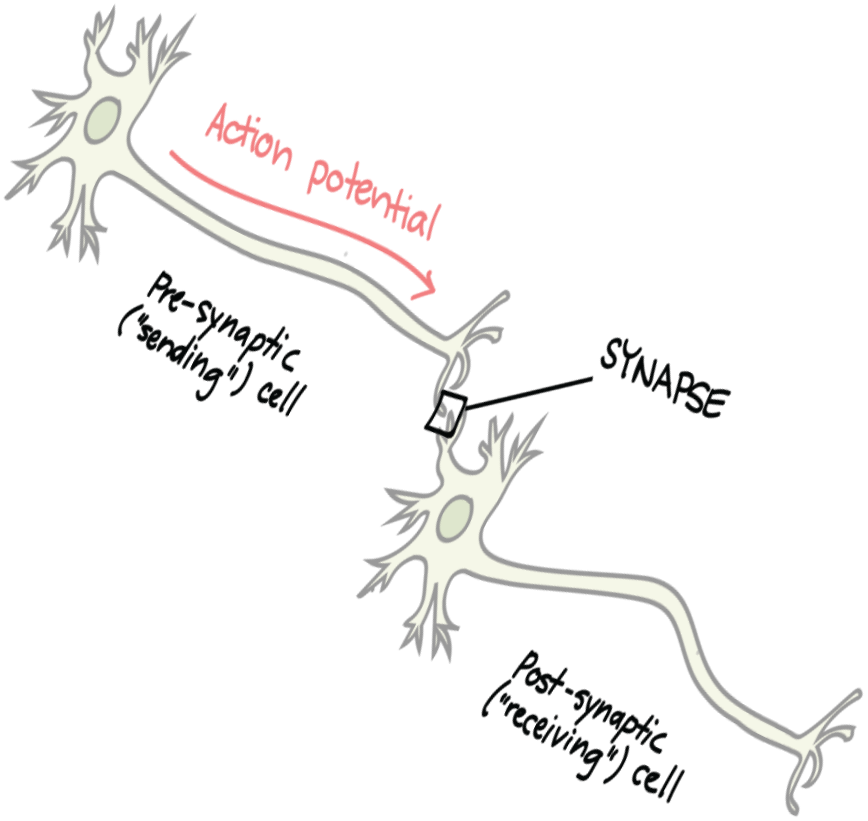
Saltatory Conduction – Action potentials appear to jump from node to node on myelinated axons



Communication between two neurons- synaptic neurotransmission

How do neurons "talk" to each other?

- The action happens at the **synapse**, the point of communication between two neurons or between a neuron and a target cell, like a muscle or a gland.
- At the synapse, the firing of an action potential in one neuron—the **pre-synaptic or** sending neuron—causes the transmission of a signal to another neuron—the **post-synaptic or** receiving neuron—making the postsynaptic neuron either more or less likely to fire its own action potential.



What are the two types of synapses?

- **Electrical synapse**
- **Chemical synapse**

At **electrical synapses**, there is a direct physical connection between the presynaptic neuron and the postsynaptic neuron through gap junctions and allows ions to flow directly from one cell into another. Gap junctions are found in Visceral smooth muscle, cardiac muscle, and the brain. **Advantages of electrical synapse is rapid conduction and coordination.**

At **chemical synapses**, there is release of neurotransmitters (chemicals) in synaptic clefts (a tiny space filled with interstitial fluid) when action potentials reaches the axon terminals at presynaptic neuron. Neurotransmitters are stored in the synaptic vesicles, and upon release into the clefts, they binds to postsynaptic neuron and depolarizes to generate next action potential at postsynaptic cell.

Neurotransmitter:–in the brain & spinal cord

- Neurotransmitters are chemicals that convey information to the target cells.

Acetylcholine (Ach)	Affects movement, learning, memory, Sleep
Dopamine (DA)	Attention, learning. regulation of movements and coordination, emotions, voluntary decision-making ability
Nor-epihephrine (NE)	Affects eating,Regulation of mood, cognition, perception, locomotion, cardiovascular functioning, and sleep.
Epinephrine	Affects metabolism of glucose, energy release during exercise.
Serotonin	Affects mood, sleep, appetite,, aggression

Glutamate	Active in areas of the brain involved in learning good emotion
GABA (Gamma-amino butyric Acid)	Facilitates neural inhibition in the central nervous system (Too much action potential)
Endorphins	Provide relief from pain and feelings of pleasure and well-being

Sample questions

Nervous system

1. System of the body which coordinates and controls its activity is known as _____

a) Organ system
b) Muscular system
c) Nervous tissue
d) Nervous system

2. Peripheral nervous system consists of the brain and spinal cord.
a) True
b) False

3. How many types of neurons are present?
a) 2
b) 1
c) 3
d) 4

4. Neurotransmission through a synapse is always from _____ to _____
a) Axon, Dendron
b) Dendron, Axon
c) Brain, spinal cord
d) Spinal cord, brain

5. Connection between axon and dendrite is called _____
a) Synapsis
b) Synapse
c) Desmosome
d) Tight junction

6. Neuroglial cells support and protect ----------
a) Nephron
b) Glands
c) Neuron
d) Muscle cells

7. Hypothalamus and thalamus are present in ---------
a) Cerebellum
b) Cerebrum
c) Diencephalon
d) Brain stem

8. Which one of the followings is the function of parasympathetic nervous system?

a) Stimulates oils and sweat glands
b) Constricts pupil
c) Acceleration of heart beat
d) Contraction of hair muscles

9. A short Gap in the myelin sheath around a nerve fiber is called _____.

a) Dendrite
b) Axon terminal
c) Nodes of Ranvier
d) None of these

10. During the propagation of a nerve impulse, the action potential results from the movement of-

a) Na+ ions from extracellular fluid to intracellular fluid
b) K+ ions from extracellular fluid to intracellular fluid
c) Na+ ions from intracellular fluid to extracellular fluid
d) K+ ions from intracellular fluid to extracellular fluid